

pempek_stats_figures

2026-03-10

This script is a general guide for making publication-quality visualizations using QIIME2-generated outputs

Research Questions

1. Does microbial community composition change over accumulated degree days?
2. Does facility influence microbial diversity or community composition?
3. Do soil and skin samples differ in their microbial diversity and community composition?

- Your research question doesn't need to be included in your R Markdown, but you should have a data analysis plan. It's easy to go down rabbit holes if you don't stick to analyses that directly address your questions. #####

loading packages

Load packages

```
library(tidyverse)
```

```
## Warning: package 'ggplot2' was built under R version 4.5.2
```

```
## Warning: package 'tibble' was built under R version 4.5.2
```

```
## Warning: package 'tidyr' was built under R version 4.5.2
```

```
## Warning: package 'readr' was built under R version 4.5.2
```

```
## Warning: package 'purrr' was built under R version 4.5.2
```

```
## Warning: package 'dplyr' was built under R version 4.5.2
```

```
## Warning: package 'lubridate' was built under R version 4.5.2
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.2.1      v readr      2.2.0
## v forcats    1.0.1      v stringr   1.6.0
## v ggplot2    4.0.3      v tibble   3.3.1
## v lubridate  1.9.5      v tidyr    1.3.2
## v purrr      1.2.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggplot2)
library(readr)
library(ggpubr)
```

```
## Warning: package 'ggpubr' was built under R version 4.5.2
```

```
library(vegan)
```

```
## Warning: package 'vegan' was built under R version 4.5.2
```

```
## Loading required package: permute
```

```
## Warning: package 'permute' was built under R version 4.5.2
```

```
library(scales)
```

```
##
## Attaching package: 'scales'
##
## The following object is masked from 'package:purrr':
##
##   discard
##
## The following object is masked from 'package:readr':
##
##   col_factor
```

```
library(lme4)
```

```
## Warning: package 'lme4' was built under R version 4.5.2
```

```
## Loading required package: Matrix
```

```
## Warning: package 'Matrix' was built under R version 4.5.2
```

```
##
## Attaching package: 'Matrix'
##
## The following objects are masked from 'package:tidyr':
```

```
##
##   expand, pack, unpack
##
## Registered S3 method overwritten by 'lme4':
##   method      from
##   na.action.merMod car
```

```
library(lmerTest)
```

```
## Warning: package 'lmerTest' was built under R version 4.5.2
```

```
##
## Attaching package: 'lmerTest'
##
## The following object is masked from 'package:lme4':
##
##   lmer
##
## The following object is masked from 'package:stats':
##
##   step
```

```
library(pbkrtest)
library(emmeans)
```

```
## Warning: package 'emmeans' was built under R version 4.5.2
```

```
## Welcome to emmeans.
## Caution: You lose important information if you filter this package's results.
## See '? untidy'
```

```
library(rstatix)
```

```
##
## Attaching package: 'rstatix'
##
## The following object is masked from 'package:stats':
##
##   filter
```

```
library(forcats)
library(scales)
```

Metadata

Import and clean metadata if needed

```

metadata <- read_tsv("/Users/dominikusraditya/Documents/ANeq505_HW/pempek/03_metadata/metadata.txt")

## Rows: 105 Columns: 20
## -- Column specification -----
## Delimiter: "\t"
## chr (13): sample, sample_type, date, age_d, age_w, sex, treatment, gel_resul...
## dbl (7): sampleid, calf_number, calf_id, day, week, weekly_fecal_counts, pen
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.

```

Alpha diversity

Import alpha diversity data from QIIME2 output

```

# Alpha diversity metrics
shannon <- read_tsv("/Users/dominikusraditya/Documents/ANeq505_HW/pempek/04_code/alpha_div/shannon.tsv")

## New names:
## Rows: 105 Columns: 2
## -- Column specification
## ----- Delimiter: "\t" dbl
## (2): ...1, shannon_entropy
## i Use 'spec()' to retrieve the full column specification for this data. i
## Specify the column types or set 'show_col_types = FALSE' to quiet this message.
## * ' -> '...1'

evenness <- read_tsv("/Users/dominikusraditya/Documents/ANeq505_HW/pempek/04_code/alpha_div/evenness.tsv")

## New names:
## Rows: 105 Columns: 2
## -- Column specification
## ----- Delimiter: "\t" dbl
## (2): ...1, pielou_evenness
## i Use 'spec()' to retrieve the full column specification for this data. i
## Specify the column types or set 'show_col_types = FALSE' to quiet this message.
## * ' -> '...1'

observed_features <- read_tsv("/Users/dominikusraditya/Documents/ANeq505_HW/pempek/04_code/alpha_div/observed_features.tsv")

## New names:
## Rows: 105 Columns: 2
## -- Column specification
## ----- Delimiter: "\t" dbl
## (2): ...1, observed_features
## i Use 'spec()' to retrieve the full column specification for this data. i
## Specify the column types or set 'show_col_types = FALSE' to quiet this message.
## * ' -> '...1'

```

```

faiths_pd <- read_tsv("/Users/dominikusraditya/Documents/ANeq505_HW/pempek/04_code/alpha_div/faith_pd.tsv")

## New names:
## Rows: 105 Columns: 2
## -- Column specification
## ----- Delimiter: "\t" dbl
## (2): ...1, faith_pd
## i Use 'spec()' to retrieve the full column specification for this data. i
## Specify the column types or set 'show_col_types = FALSE' to quiet this message.
## * ' -> '...1'

```

Tidy metrics and add to metadata

```

# Shannon
shannon <- shannon %>%
  rename(sampleid = 1) #rename 1st column to to match metadata

# Evenness
evenness <- evenness %>%
  rename(sampleid = 1) #rename 1st column to to match metadata

# Observed Features
observed_features <- observed_features %>%
  rename(sampleid = 1) #rename 1st column to to match metadata

# Faiths PD
faiths_pd <- faiths_pd %>%
  rename(sampleid = 1) #rename 1st column to to match metadata

# Add to metadata
metadata <- metadata %>%
  left_join(shannon, by = "sampleid") %>%
  left_join(evenness, by = "sampleid") %>%
  left_join(observed_features, by = "sampleid") %>%
  left_join(faiths_pd, by = "sampleid")

```

Tidy metadata with metrics

```

metadata <- metadata %>%
  drop_na(shannon_entropy, pielou_evenness, observed_features, faith_pd)

```

Linear Mixed Effects Models

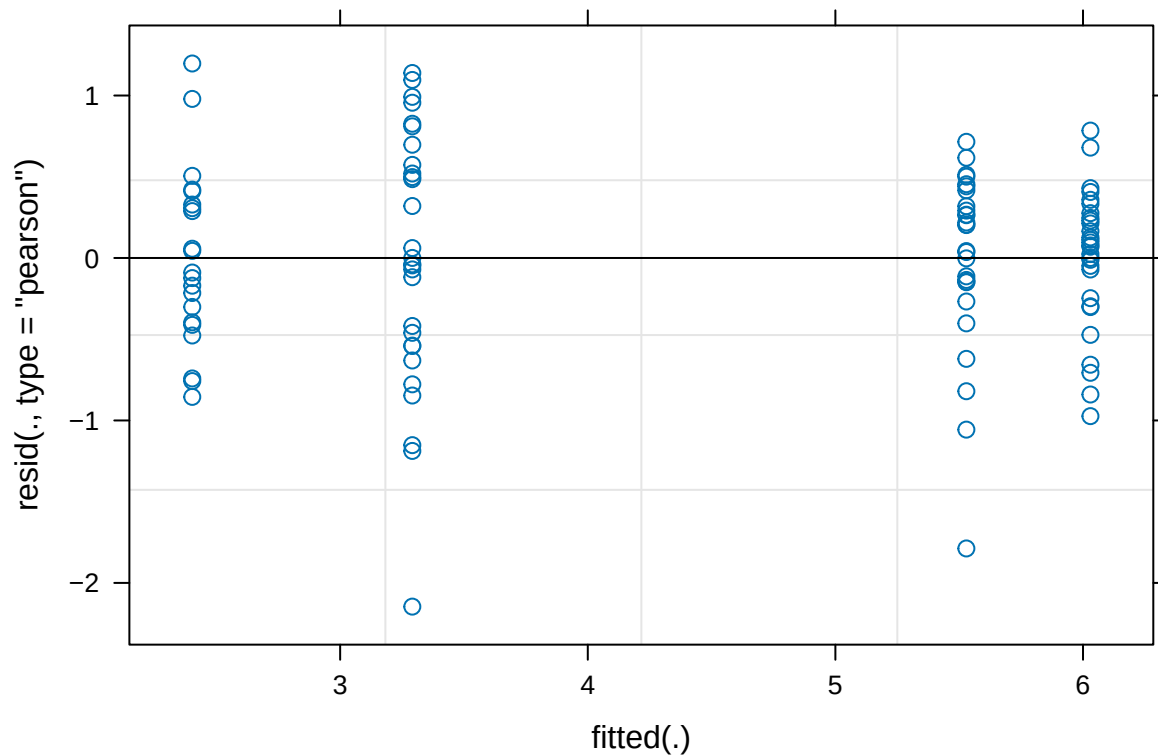
Time (question 1)

Shannon

```
# Build model
model_age_d_shannon <- lmer(shannon_entropy ~ age_d + (1|calf_id), data = metadata) # I name all my mod
```

```
## boundary (singular) fit: see help('isSingular')
```

```
# Residual diagnostic plot, check assumptions
plot(model_age_d_shannon)
```



```
# look at estimated variance components
summary(model_age_d_shannon)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: shannon_entropy ~ age_d + (1 | calf_id)
## Data: metadata
##
## REML criterion at convergence: 196.9
##
## Scaled residuals:
## Min 1Q Median 3Q Max
## -3.5696 -0.5025 0.0732 0.6763 1.9916
##
## Random effects:
## Groups Name Variance Std.Dev.
## calf_id (Intercept) 3.164e-18 1.779e-09
## Residual 3.614e-01 6.012e-01
## Number of obs: 105, groups: calf_id, 30
##
```

```
## Fixed effects:
##           Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  2.4027    0.1312 101.0000  18.316 < 2e-16 ***
## age_dday 35   3.1263    0.1749 101.0000  17.873 < 2e-16 ***
## age_dday 5    0.8887    0.1735 101.0000   5.121 1.46e-06 ***
## age_dday 63   3.6273    0.1723 101.0000  21.058 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr) ag_d35 ag_dd5
## age_dday 35 -0.750
## age_dday 5  -0.756  0.567
## age_dday 63 -0.762  0.571  0.576
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```
# residual diagnostic plot, check assumptions
anova(model_age_d_shannon)
```

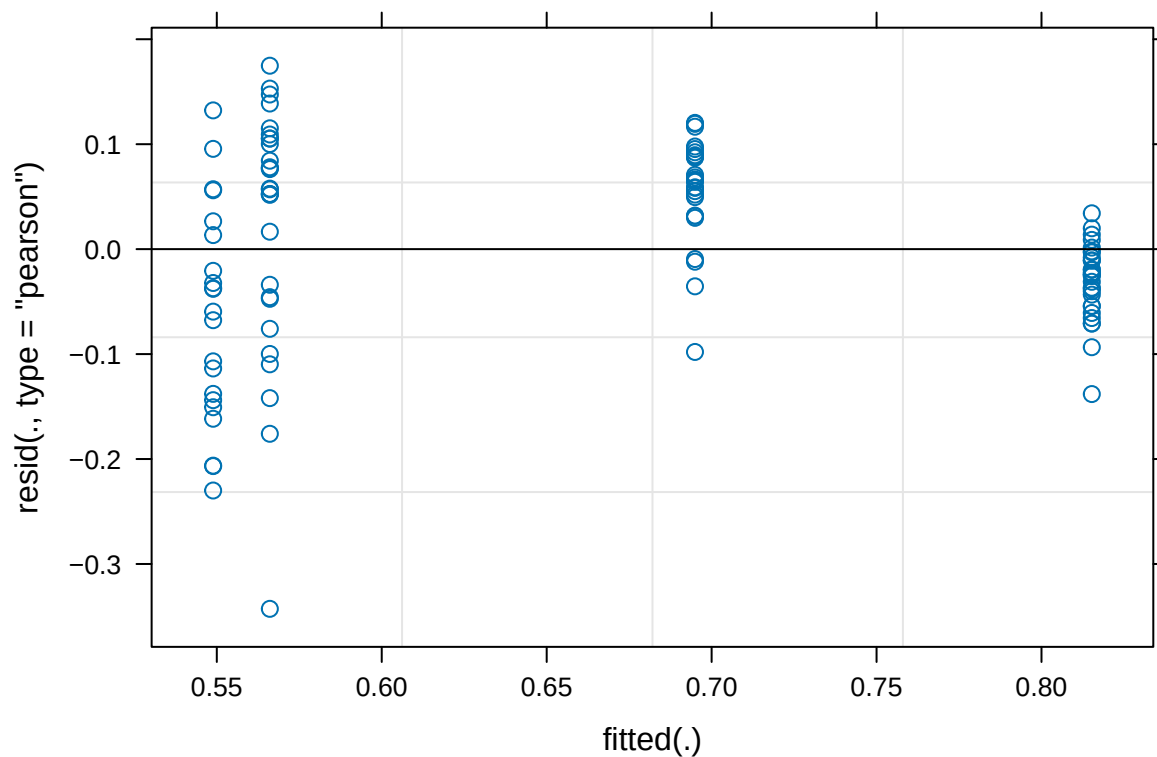
```
## Type III Analysis of Variance Table with Satterthwaite's method
##           Sum Sq Mean Sq NumDF DenDF F value    Pr(>F)
## age_d 229.43  76.478      3    101  211.62 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Evenness

```
# Build model
model_day_evenness <- lmer(pielou_evenness ~ day + (1|calf_id), data = metadata) # I name all my models
```

```
## boundary (singular) fit: see help('isSingular')
```

```
# Residual diagnostic plot, check assumptions
plot(model_day_evenness)
```



```
# look at estimated variance components
summary(model_day_evenness)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: pielou_evenness ~ day + (1 | calf_id)
## Data: metadata
##
## REML criterion at convergence: -179.3
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.6504 -0.5041  0.0157  0.7282  1.8617
##
## Random effects:
## Groups Name Variance Std.Dev.
## calf_id (Intercept) 0.000000 0.00000
## Residual 0.008815 0.09389
## Number of obs: 105, groups: calf_id, 30
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept) 5.446e-01 1.366e-02 1.030e+02 39.86 <2e-16 ***
## day         4.296e-03 3.628e-04 1.030e+02 11.84 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr)
```

```
## day -0.742
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```
# residual diagnostic plot, check assumptions
anova(model_day_evenness)
```

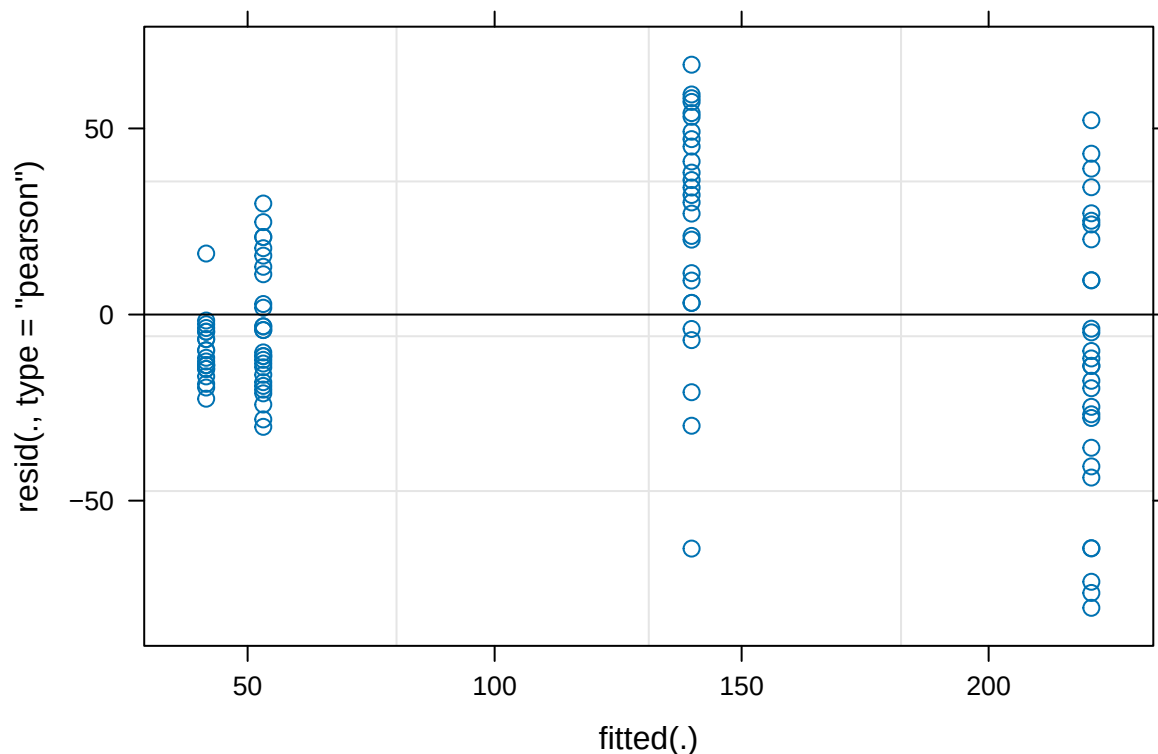
```
## Type III Analysis of Variance Table with Satterthwaite's method
##      Sum Sq Mean Sq NumDF DenDF F value    Pr(>F)
## day  1.236   1.236     1   103  140.21 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Observed Features

```
# Build model
model_day_obfeatures <- lmer(observed_features ~ day + (1|calf_id), data = metadata) # I name all my mo
```

```
## boundary (singular) fit: see help('isSingular')
```

```
# Residual diagnostic plot, check assumptions
plot(model_day_obfeatures)
```



```
# look at estimated variance components
summary(model_day_obfeatures)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: observed_features ~ day + (1 | calf_id)
## Data: metadata
##
## REML criterion at convergence: 1012.8
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.5747 -0.5427 -0.1506  0.6807  2.1936
##
## Random effects:
## Groups Name Variance Std.Dev.
## calf_id (Intercept) 1.021e-16 1.011e-08
## Residual 9.365e+02 3.060e+01
## Number of obs: 105, groups: calf_id, 30
##
## Fixed effects:
## Estimate Std. Error df t value Pr(>|t|)
## (Intercept) 38.7190 4.4532 103.0000 8.695 5.82e-14 ***
## day 2.8900 0.1183 103.0000 24.439 < 2e-16 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
## (Intr)
## day -0.742
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```
# residual diagnostic plot, check assumptions
anova(model_day_obfeatures)
```

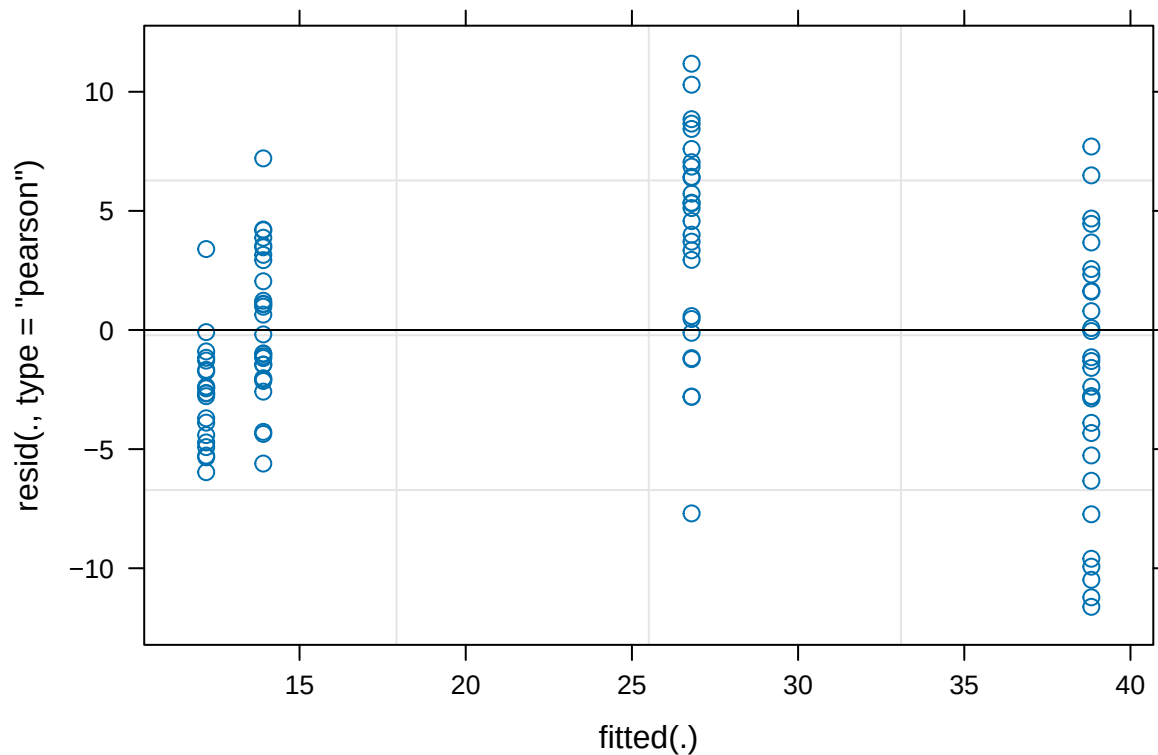
```
## Type III Analysis of Variance Table with Satterthwaite's method
## Sum Sq Mean Sq NumDF DenDF F value Pr(>F)
## day 559346 559346 1 103 597.27 < 2.2e-16 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Faiths PD

```
# Build model
model_day_faithpd <- lmer(faith_pd ~ day + (1|calf_id), data = metadata) # I name all my models with th
```

```
## boundary (singular) fit: see help('isSingular')
```

```
# Residual diagnostic plot, check assumptions
plot(model_day_faithpd)
```



```
# look at estimated variance components
summary(model_day_faithpd)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: faith_pd ~ day + (1 | calf_id)
## Data: metadata
##
## REML criterion at convergence: 631.9
##
## Scaled residuals:
##   Min       1Q   Median       3Q      Max
## -2.4130 -0.5770 -0.2035  0.7344  2.3213
##
## Random effects:
## Groups Name Variance Std.Dev.
## calf_id (Intercept) 0.00 0.000
## Residual 23.19 4.815
## Number of obs: 105, groups: calf_id, 30
##
## Fixed effects:
## Estimate Std. Error df t value Pr(>|t|)
## (Intercept) 11.76155 0.70073 103.00000 16.79 <2e-16 ***
## day 0.42955 0.01861 103.00000 23.08 <2e-16 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
## (Intr)
```

```
## day -0.742
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```
# residual diagnostic plot, check assumptions
anova(model_day_faithpd)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##      Sum Sq Mean Sq NumDF DenDF F value    Pr(>F)
## day  12356   12356      1   103  532.86 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

day split into 3

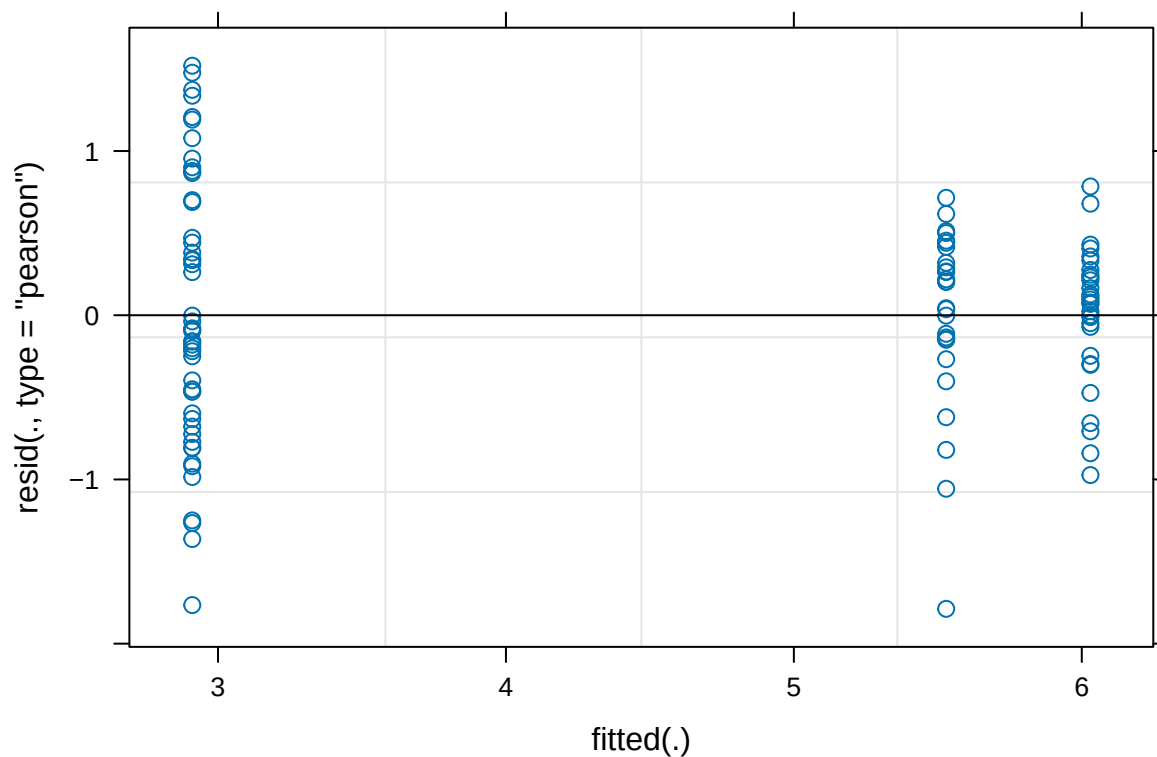
Shannon

```
# Add metadata column
metadata <- metadata %>%
  mutate(day_split = cut(
    day,
    breaks = 3,
    labels = c("1-5", "5-35", "35-63"),
    include.lowest = TRUE
  ))

# Stats
# Build model
model_daysplit_shannon <- lmer(shannon_entropy ~ day_split + (1|calf_id), data = metadata)
```

```
## boundary (singular) fit: see help('isSingular')
```

```
# Residual diagnostic plot, check assumptions
plot(model_daysplit_shannon)
```



```
# look at estimated variance components
summary(model_daysplit_shannon)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: shannon_entropy ~ day_split + (1 | calf_id)
## Data: metadata
##
## REML criterion at convergence: 218.7
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.66289 -0.59924  0.03602  0.56797  2.26326
##
## Random effects:
## Groups Name Variance Std.Dev.
## calf_id (Intercept) 0.0000  0.0000
## Residual          0.4508  0.6714
## Number of obs: 105, groups: calf_id, 30
##
## Fixed effects:
##              Estimate Std. Error    df t value Pr(>|t|)
## (Intercept)    2.91056    0.09591 102.00000   30.35 <2e-16 ***
## day_split5-35    2.61845    0.16092 102.00000   16.27 <2e-16 ***
## day_split35-63    3.11947    0.15730 102.00000   19.83 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
```

```
##          (Intr) d_5-35
## dy_splt5-35 -0.596
## dy_spl35-63 -0.610  0.363
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```
# residual diagnostic plot, check assumptions
anova(model_daysplit_shannon)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##          Sum Sq Mean Sq NumDF DenDF F value    Pr(>F)
## day_split 219.96  109.98     2   102  243.98 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# pairwise comparisons
emmeans(model_daysplit_shannon, pairwise ~ day_split)
```

```
## $emmeans
##   day_split emmean      SE    df lower.CL upper.CL
## 1-5         2.91 0.0963  80.9     2.72     3.10
## 5-35         5.53 0.1290 102.0     5.27     5.79
## 35-63         6.03 0.1250 102.0     5.78     6.28
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## $contrasts
##   contrast      estimate      SE    df t.ratio p.value
## (1-5) - (5-35)    -2.618 0.161  79.5 -16.233 <0.0001
## (1-5) - (35-63)   -3.119 0.158  78.1 -19.796 <0.0001
## (5-35) - (35-63)  -0.501 0.180  75.5  -2.789  0.0182
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: tukey method for comparing a family of 3 estimates
```

day figure (volatility plot)

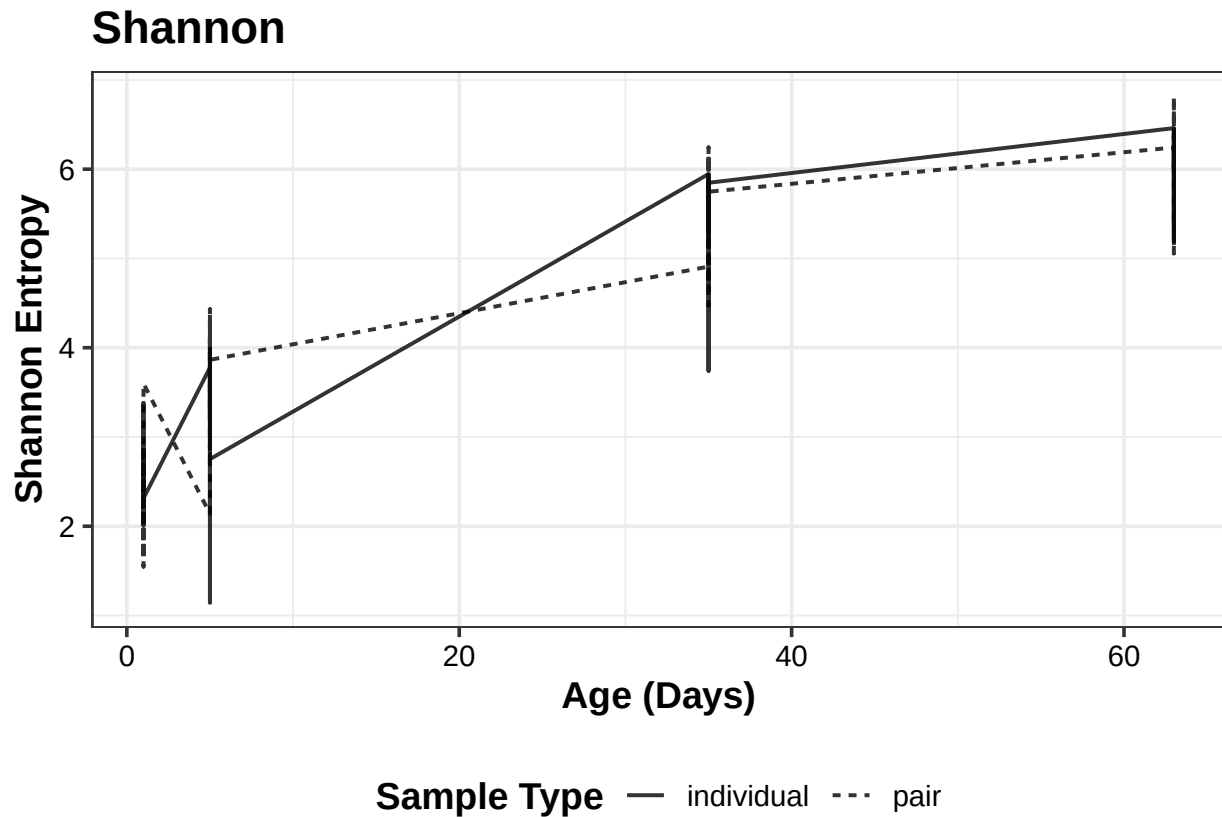
```
# Shannon
day_shannon <- ggplot(metadata, aes(x = day, y = shannon_entropy, linetype = treatment)) +
  geom_line(linewidth = 0.7, alpha = 0.8) +
  theme_bw(base_size = 14) +
  labs(
    x = "Age (Days)",
    y = "Shannon Entropy",
    linetype = "Sample Type",
    title = "Shannon"
  ) +
  theme(
    axis.title = element_text(face = "bold"),
```

```

axis.text = element_text(color = "black"),
legend.title = element_text(face = "bold"),
legend.position = "bottom",
legend.background = element_rect(fill = "transparent"),
plot.title = element_text(face = "bold")
)

```

day_shannon



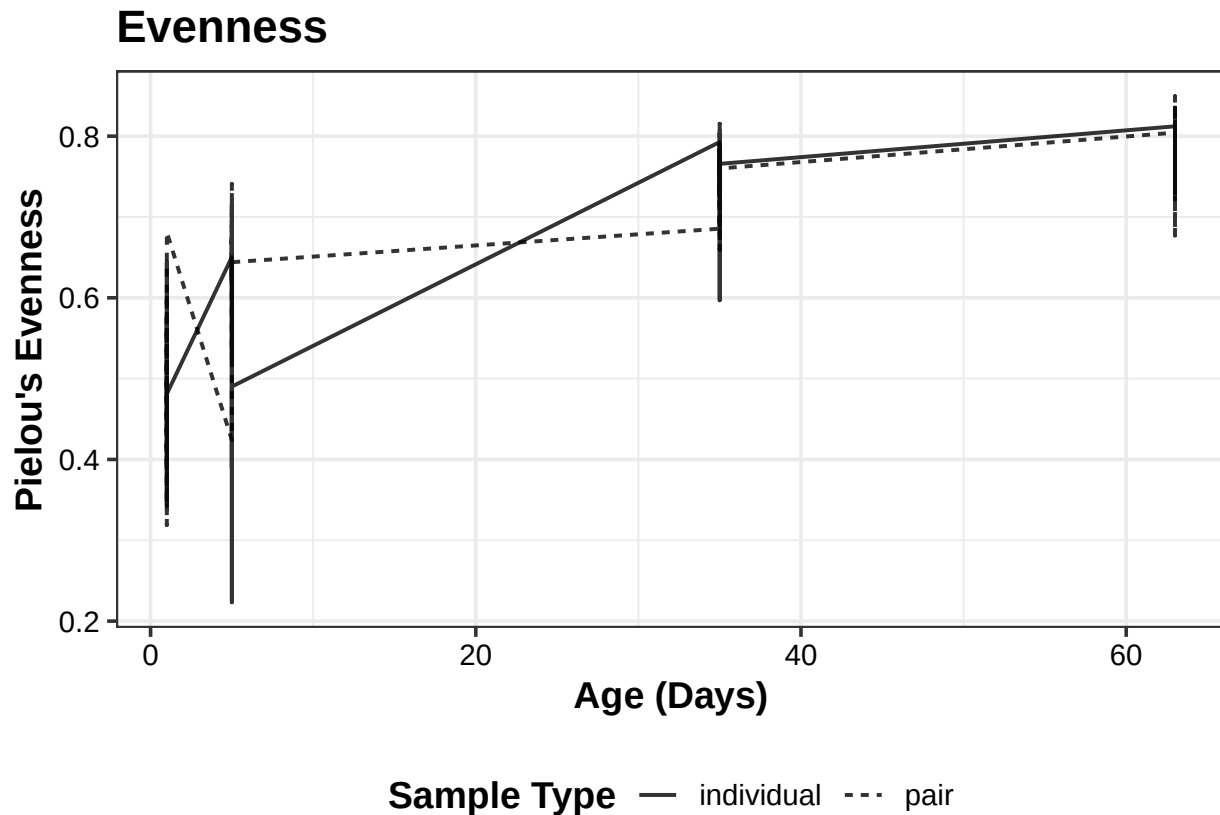
```

# Evenness
day_evenness <- ggplot(metadata, aes(x = day, y = pielou_evenness, linetype = treatment)) +
  geom_line(size = 0.7, alpha = 0.8) +
  theme_bw(base_size = 14) +
  labs(
    x = "Age (Days)",
    y = "Pielou's Evenness",
    linetype = "Sample Type",
    title = "Evenness"
  ) +
  theme(
    axis.title = element_text(face = "bold"),
    axis.text = element_text(color = "black"),
    legend.title = element_text(face = "bold"),
    legend.position = "bottom",
    legend.background = element_rect(fill = "transparent"),
    plot.title = element_text(face = "bold")
  )

```

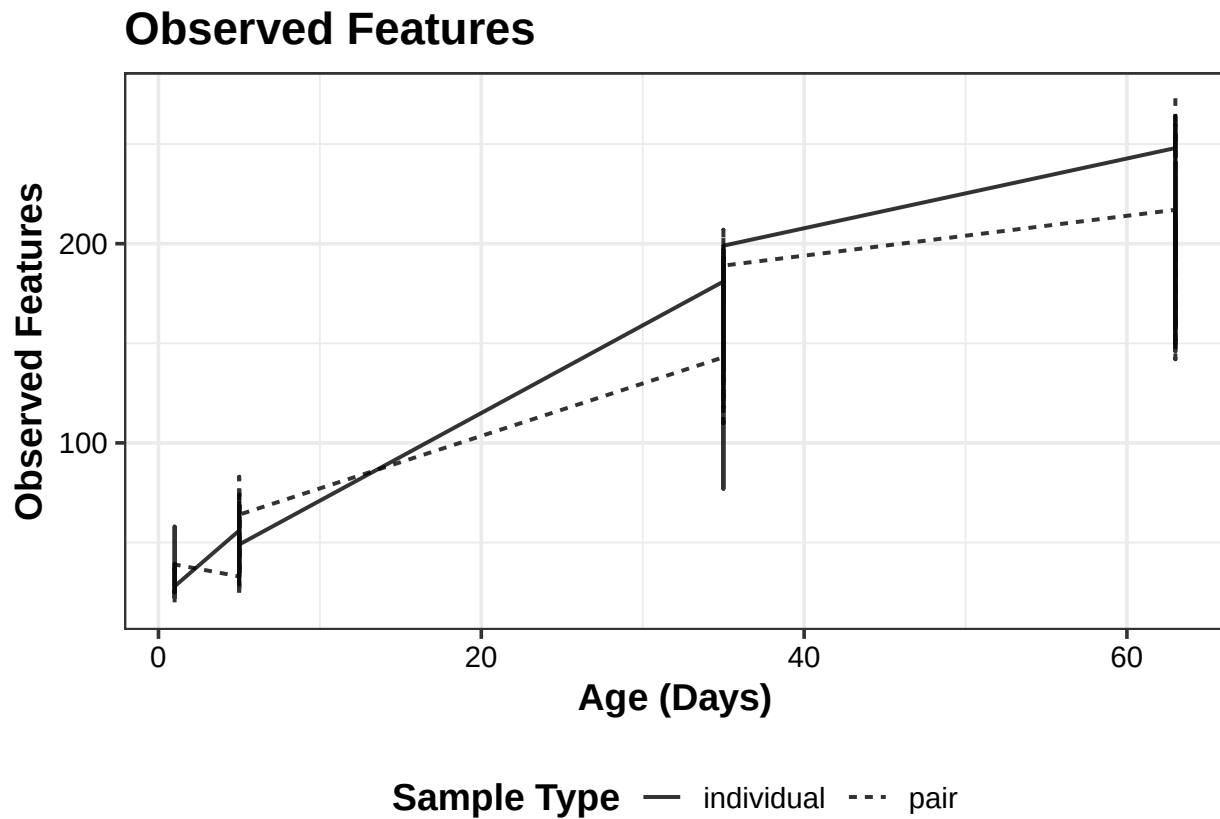
```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once per session.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

```
day_evenness
```

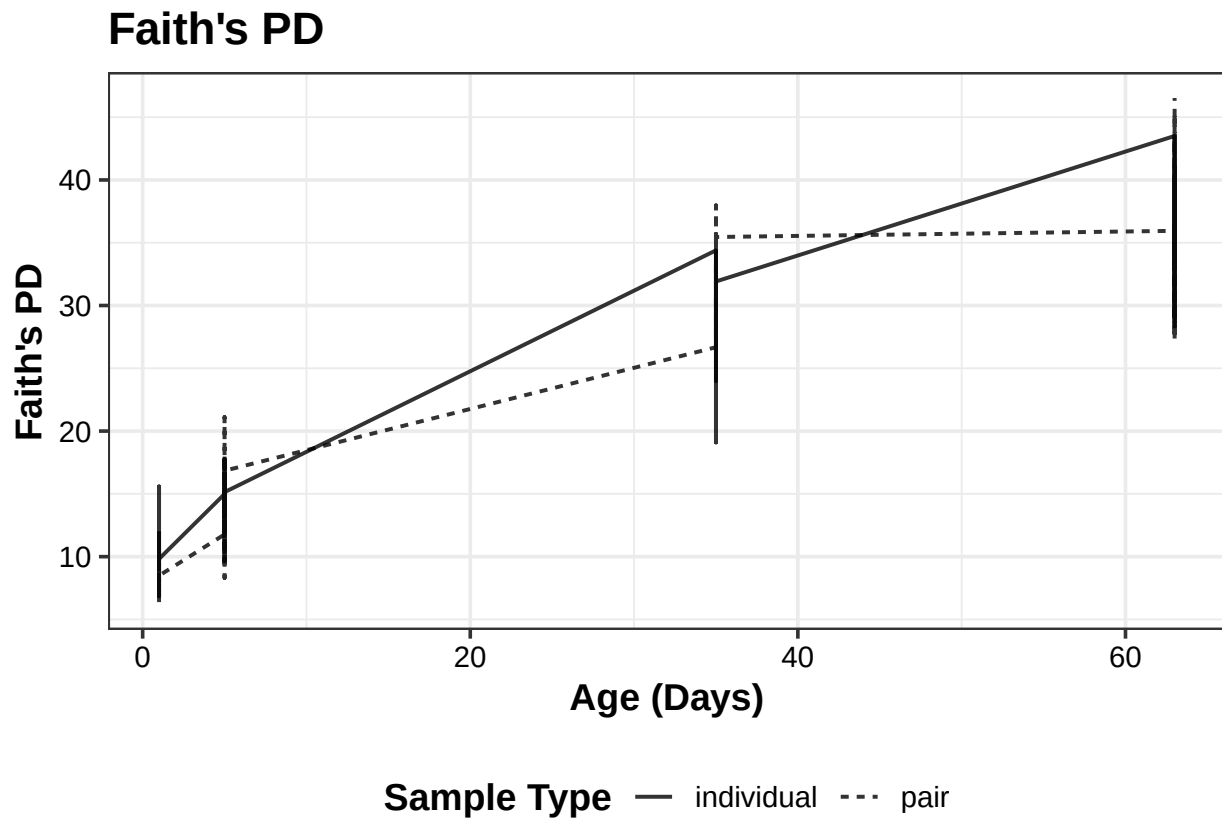


```
# Observed Features
day_obfeatures <- ggplot(metadata, aes(x = day, y = observed_features, linetype = treatment)) +
  geom_line(size = 0.7, alpha = 0.8) +
  theme_bw(base_size = 14) +
  labs(
    x = "Age (Days)",
    y = "Observed Features",
    linetype = "Sample Type",
    title = "Observed Features"
  ) +
  theme(
    axis.title = element_text(face = "bold"),
    axis.text = element_text(color = "black"),
    legend.title = element_text(face = "bold"),
    legend.position = "bottom",
    legend.background = element_rect(fill = "transparent"),
    plot.title = element_text(face = "bold")
  )
```

day_obfeatures

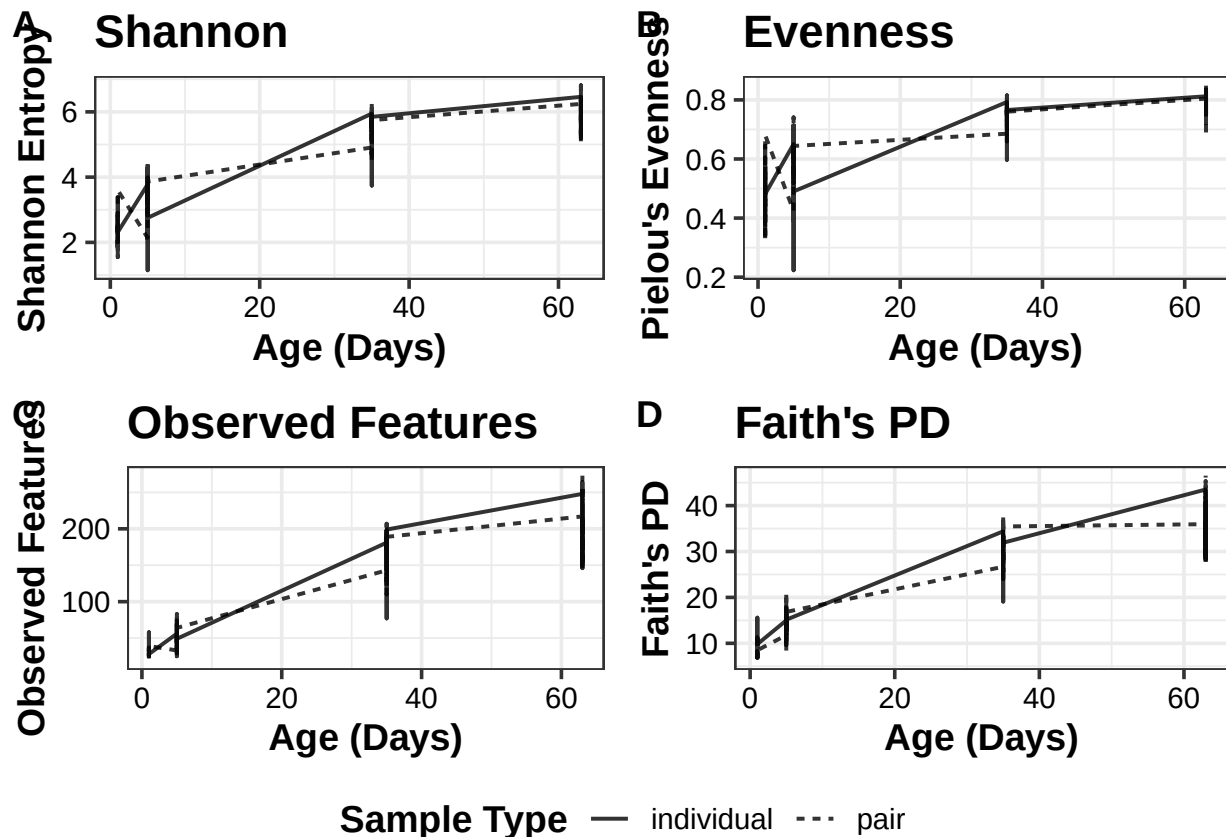


```
# Faith's PD
day_faithpd <- ggplot(metadata, aes(x = day, y = faith_pd, linetype = treatment)) +
  geom_line(size = 0.7, alpha = 0.8) +
  theme_bw(base_size = 14) +
  labs(
    x = "Age (Days)",
    y = "Faith's PD",
    linetype = "Sample Type",
    title = "Faith's PD"
  ) +
  theme(
    axis.title = element_text(face = "bold"),
    axis.text = element_text(color = "black"),
    legend.title = element_text(face = "bold"),
    legend.position = "bottom",
    legend.background = element_rect(fill = "transparent"),
    plot.title = element_text(face = "bold")
  )
day_faithpd
```



```
# Combine the plots
alpha_div_day <- ggarrange(day_shannon, day_evenness,
                           day_obfeatures, day_faithpd,
                           ncol = 2, nrow = 2,
                           labels = c("A", "B", "C", "D"),
                           common.legend = TRUE,
                           legend = "bottom")

alpha_div_day
```



```
# Save the figure
ggsave(alpha_div_day, filename = "../05_figures/alpha_div_add0c.jpeg", device = "jpeg", dpi = 300, width = 1000, height = 1000)

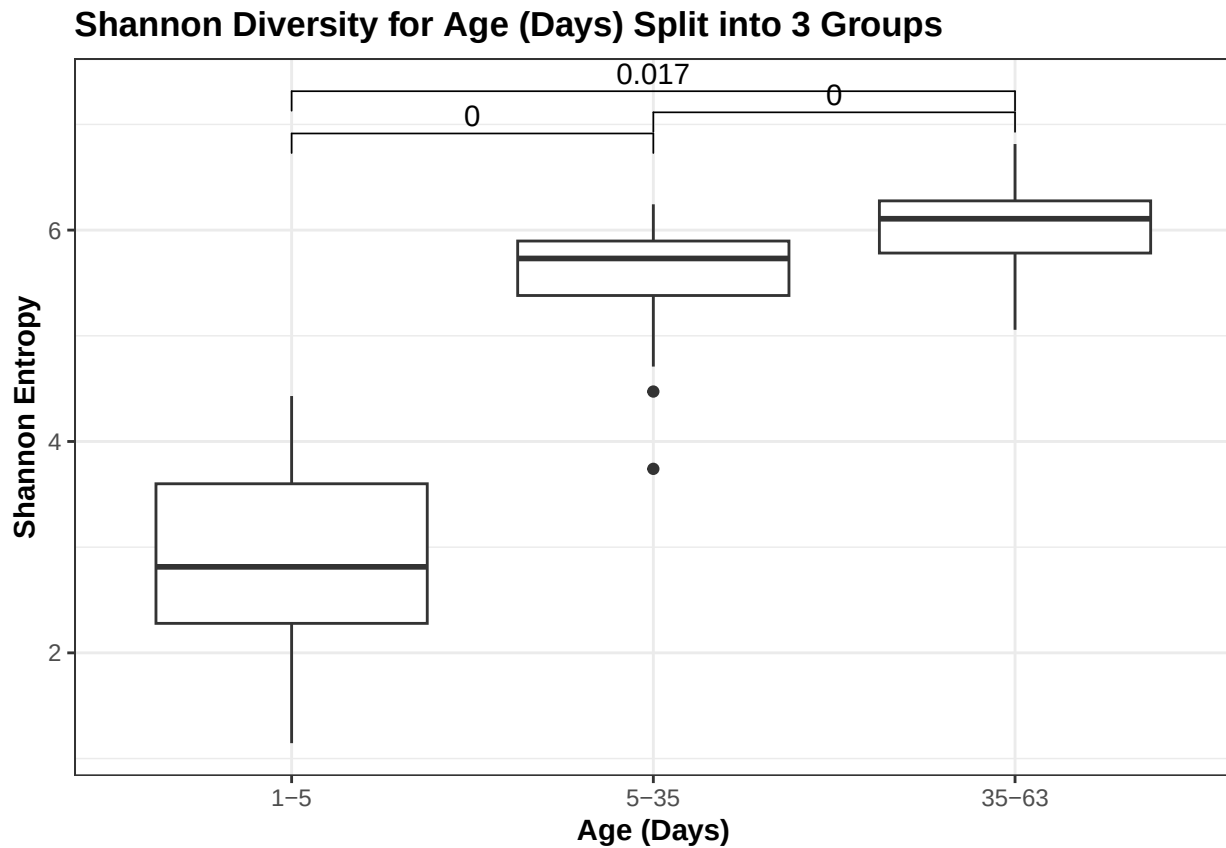
model_daysplit_shannon <- lmer(shannon_entropy ~ day_split + (1|treatment), data = metadata)

pvals <- pairs(emmeans(model_daysplit_shannon, ~ day_split)) |> as.data.frame()

# make table for stat_pvalue_manual with rounded p-values
pvals_table <- data.frame(
  group1 = c("1-5", "5-35", "35-63"),
  group2 = c("5-35", "35-63", "1-5"),
  p = round(pvals$p.value, 3), # round to 3 decimals
  y.position = max(metadata$shannon_entropy) + c(0.1, 0.3, 0.5)
)

# plot
day_split_shannon <- ggplot(metadata, aes(x = day_split, y = shannon_entropy)) +
  geom_boxplot() +
  stat_pvalue_manual(pvals_table, label = "p") +
  theme_bw() +
  theme(
    axis.title = element_text(face = "bold"),
    plot.title = element_text(face = "bold")
  ) +
  labs(x = "Age (Days)",
       y = "Shannon Entropy",
       title = "Shannon Diversity for Age (Days) Split into 3 Groups")
```

```
day_split_shannon
```



```
# Save the figure
```

```
ggsave(day_split_shannon, filename = "../05_figures/day_split_shannon.jpeg", device = "jpeg", dpi = 300)
```

Day + Treatment (question 2)

Shannon

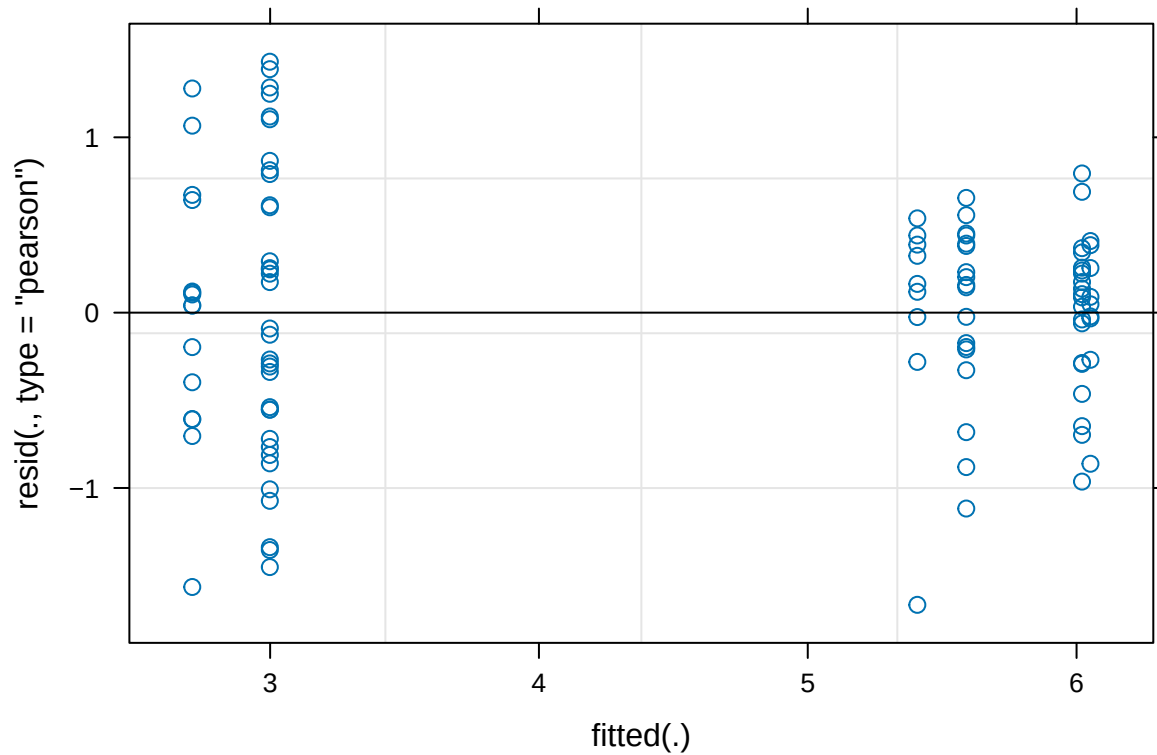
```
# Build model
```

```
model_age_treatment_shannon <- lmer(shannon_entropy ~ treatment*day_split + (1|calf_id), data = metadata)
```

```
## boundary (singular) fit: see help('isSingular')
```

```
# Residual diagnostic plot, check assumptions
```

```
plot(model_age_treatment_shannon)
```



```
# look at estimated variance components
summary(model_age_treatment_shannon)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: shannon_entropy ~ treatment * day_split + (1 | calf_id)
## Data: metadata
##
## REML criterion at convergence: 219.2
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.4743 -0.5024  0.1337  0.5716  2.1250
##
## Random effects:
## Groups Name Variance Std.Dev.
## calf_id (Intercept) 0.0000  0.0000
## Residual          0.4536  0.6735
## Number of obs: 105, groups: calf_id, 30
##
## Fixed effects:
##              Estimate Std. Error    df t value Pr(>|t|)
## (Intercept)      2.7103    0.1739 99.0000  15.586 < 2e-16 ***
## treatmentpair      0.2886    0.2088 99.0000   1.382  0.170
## day_split5-35      2.6973    0.2840 99.0000   9.498 1.36e-15 ***
## day_split35-63      3.3413    0.2840 99.0000  11.766 < 2e-16 ***
## treatmentpair:day_split5-35 -0.1065    0.3452 99.0000  -0.309  0.758
## treatmentpair:day_split35-63 -0.3199    0.3416 99.0000  -0.937  0.351
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr) trtmnt d_5-35 d_35-6 t:_5-3
## treatmentpr -0.833
## dy_splt5-35 -0.612  0.510
## dy_spl35-63 -0.612  0.510  0.375
## trtmn:_5-35  0.504 -0.605 -0.823 -0.308
## trtm:_35-63  0.509 -0.611 -0.312 -0.831  0.370
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')

# residual diagnostic plot, check assumptions
anova(model_age_treatment_shannon)

## Type III Analysis of Variance Table with Satterthwaite's method
##              Sum Sq Mean Sq NumDF DenDF  F value Pr(>F)
## treatment          0.455    0.455     1    99   1.0038 0.3188
## day_split        194.769   97.385     2    99 214.6904 <2e-16 ***
## treatment:day_split  0.399    0.199     2    99   0.4394 0.6457
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

# pairwise comparisons
emmeans(model_age_treatment_shannon, pairwise ~ treatment|day_split)

## $emmeans
## day_split = 1-5:
##   treatment emmean    SE    df lower.CL upper.CL
## individual   2.71 0.175 78.7     2.36     3.06
## pair         3.00 0.116 77.8     2.77     3.23
##
## day_split = 5-35:
##   treatment emmean    SE    df lower.CL upper.CL
## individual   5.41 0.225 99.0     4.96     5.85
## pair         5.59 0.159 99.0     5.27     5.91
##
## day_split = 35-63:
##   treatment emmean    SE    df lower.CL upper.CL
## individual   6.05 0.225 99.0     5.61     6.50
## pair         6.02 0.151 99.0     5.72     6.32
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## $contrasts
## day_split = 1-5:
##   contrast      estimate    SE    df t.ratio p.value
## individual - pair -0.2886 0.210 78.4   -1.376  0.1728
##
## day_split = 5-35:
##   contrast      estimate    SE    df t.ratio p.value
## individual - pair -0.1821 0.275 99.0   -0.661  0.5101
```

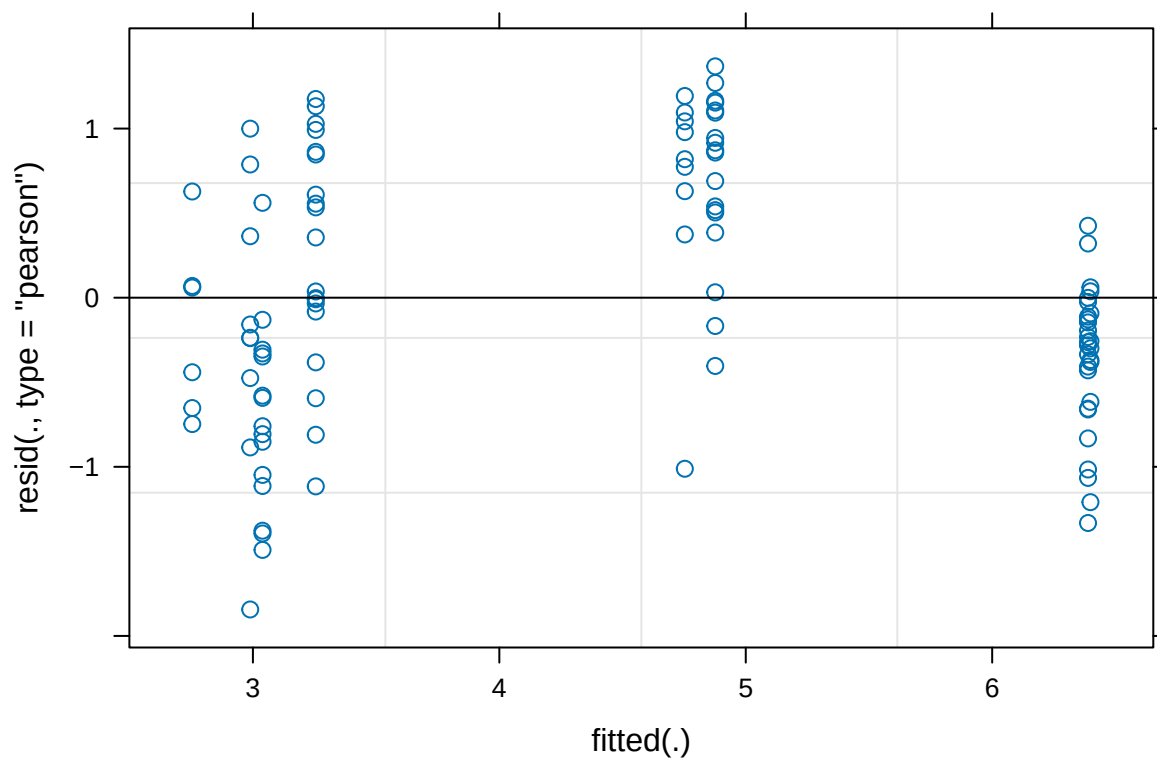
```
##
## day_split = 35-63:
## contrast      estimate    SE   df t.ratio p.value
## individual - pair  0.0313 0.271 99.0   0.116  0.9081
##
## Degrees-of-freedom method: kenward-roger
```

Really quick lets just look when we don't split day

```
# Build model
model_day_treatment_shannon2 <- lmer(shannon_entropy ~ treatment*day + (1|calf_id), data = metadata)
```

```
## boundary (singular) fit: see help('isSingular')
```

```
# Residual diagnostic plot, check assumptions
plot(model_day_treatment_shannon2)
```



```
# look at estimated variance components
summary(model_day_treatment_shannon2)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: shannon_entropy ~ treatment * day + (1 | calf_id)
## Data: metadata
##
## REML criterion at convergence: 260.8
##
## Scaled residuals:
```

```
##      Min      1Q  Median      3Q      Max
## -2.4116 -0.6217 -0.1213  0.8215  1.7895
##
## Random effects:
## Groups   Name            Variance Std.Dev.
## calf_id  (Intercept)  0.0000    0.0000
## Residual                0.5844    0.7645
## Number of obs: 105, groups:  calf_id, 30
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    2.695185   0.200965 101.000000   13.411  <2e-16 ***
## treatmentpair    0.289854   0.241307 101.000000    1.201    0.232
## day              0.058786   0.005326 101.000000   11.037  <2e-16 ***
## treatmentpair:day -0.004756   0.006401 101.000000   -0.743    0.459
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) trtmnt day
## treatmentpr -0.833
## day          -0.749  0.624
## trtmntpr:dy  0.624 -0.746 -0.832
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```
# residual diagnostic plot, check assumptions
anova(model_day_treatment_shannon2)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##              Sum Sq Mean Sq NumDF DenDF  F value Pr(>F)
## treatment      0.843   0.843     1   101   1.4428 0.2325
## day            181.523 181.523     1   101 310.6165 <2e-16 ***
## treatment:day   0.323   0.323     1   101   0.5519 0.4593
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# pairwise comparisons
emmeans(model_day_treatment_shannon2, pairwise ~ treatment)
```

```
## NOTE: Results may be misleading due to involvement in interactions
```

```
## $emmeans
## treatment emmean      SE df lower.CL upper.CL
## individual  4.34 0.1340 29.0     4.06     4.61
## pair        4.49 0.0903 25.8     4.31     4.68
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## $contrasts
## contrast      estimate      SE df t.ratio p.value
```

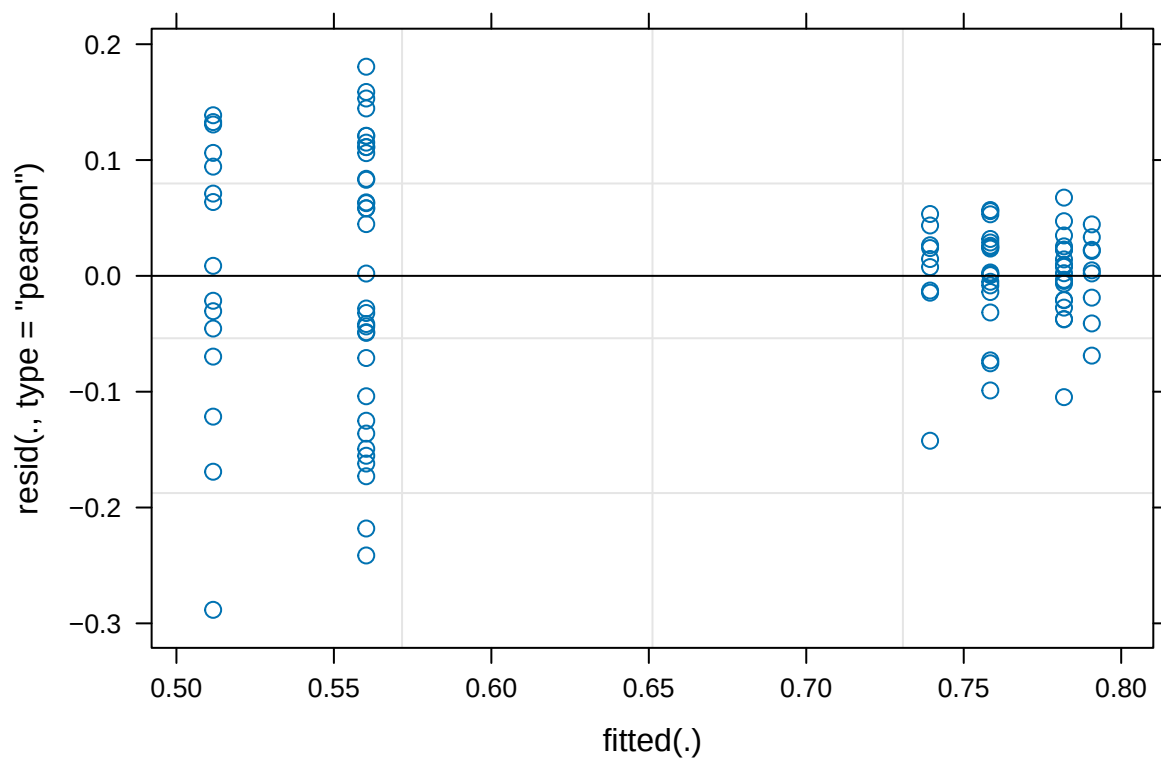
```
## individual - pair    -0.157 0.161 27.9  -0.974  0.3385
##
## Degrees-of-freedom method: kenward-roger
```

Evenness

```
# Build model
model_day_treatment_evenness <- lmer(pielou_evenness ~ treatment*day_split + (1|calf_id), data = metadata)
```

```
## boundary (singular) fit: see help('isSingular')
```

```
# Residual diagnostic plot, check assumptions
plot(model_day_treatment_evenness)
```



```
# look at estimated variance components
summary(model_day_treatment_evenness)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: pielou_evenness ~ treatment * day_split + (1 | calf_id)
## Data: metadata
##
## REML criterion at convergence: -180
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.2147 -0.4188  0.0847  0.5957  2.0135
##
```

```
## Random effects:
## Groups Name Variance Std.Dev.
## calf_id (Intercept) 0.000000 0.00000
## Residual 0.008044 0.08969
## Number of obs: 105, groups: calf_id, 30
##
## Fixed effects:
## Estimate Std. Error df t value Pr(>|t|)
## (Intercept) 0.51166 0.02316 99.00000 22.094 < 2e-16 ***
## treatmentpair 0.04862 0.02780 99.00000 1.749 0.0834 .
## day_split5-35 0.22765 0.03782 99.00000 6.020 2.97e-08 ***
## day_split35-63 0.27901 0.03782 99.00000 7.378 5.03e-11 ***
## treatmentpair:day_split5-35 -0.02949 0.04597 99.00000 -0.641 0.5228
## treatmentpair:day_split35-63 -0.05744 0.04549 99.00000 -1.263 0.2096
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
## (Intr) trtmnt d_5-35 d_35-6 t:_5-3
## treatmentpr -0.833
## dy_splt5-35 -0.612 0.510
## dy_spl35-63 -0.612 0.510 0.375
## trtmn:_5-35 0.504 -0.605 -0.823 -0.308
## trtm:_35-63 0.509 -0.611 -0.312 -0.831 0.370
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```
# residual diagnostic plot, check assumptions
anova(model_day_treatment_evenness)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
## Sum Sq Mean Sq NumDF DenDF F value Pr(>F)
## treatment 0.00819 0.00819 1 99 1.0182 0.3154
## day_split 1.22569 0.61284 2 99 76.1822 <2e-16 ***
## treatment:day_split 0.01311 0.00656 2 99 0.8151 0.4455
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

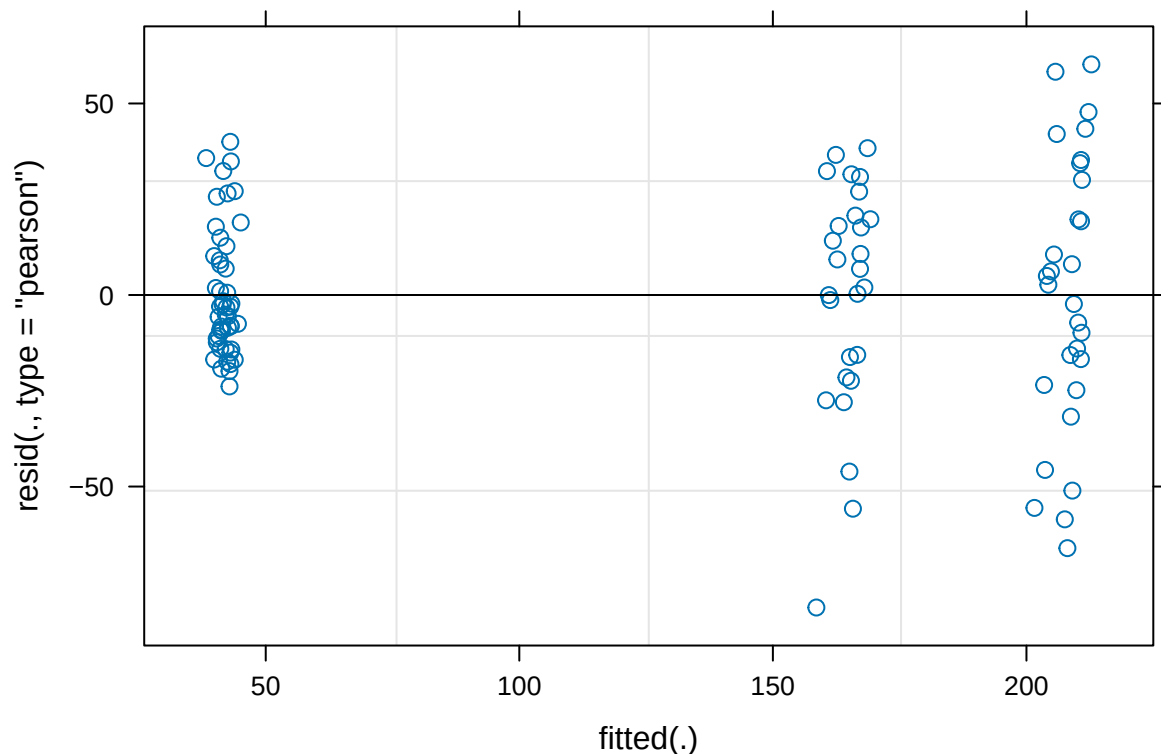
```
# pairwise comparisons
emmeans(model_day_treatment_evenness, pairwise ~ treatment|day_split)
```

```
## $emmeans
## day_split = 1-5:
## treatment emmean SE df lower.CL upper.CL
## individual 0.512 0.0233 78.7 0.465 0.558
## pair 0.560 0.0154 77.8 0.530 0.591
##
## day_split = 5-35:
## treatment emmean SE df lower.CL upper.CL
## individual 0.739 0.0299 99.0 0.680 0.799
## pair 0.758 0.0212 99.0 0.716 0.800
##
## day_split = 35-63:
```

```
## treatment emmean SE df lower.CL upper.CL
## individual 0.791 0.0299 99.0 0.731 0.850
## pair 0.782 0.0201 99.0 0.742 0.822
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## $contrasts
## day_split = 1-5:
## contrast estimate SE df t.ratio p.value
## individual - pair -0.04862 0.0279 78.4 -1.740 0.0857
##
## day_split = 5-35:
## contrast estimate SE df t.ratio p.value
## individual - pair -0.01913 0.0367 99.0 -0.522 0.6031
##
## day_split = 35-63:
## contrast estimate SE df t.ratio p.value
## individual - pair 0.00883 0.0360 99.0 0.245 0.8070
##
## Degrees-of-freedom method: kenward-roger
```

Observed Features

```
# Build model
model_day_treatment_obfeatures <- lmer(observed_features ~ treatment*day_split + (1|calf_id), data = me
# Residual diagnostic plot, check assumptions
plot(model_day_treatment_obfeatures)
```



```
# look at estimated variance components
summary(model_day_treatment_obfeatures)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: observed_features ~ treatment * day_split + (1 | calf_id)
## Data: metadata
##
## REML criterion at convergence: 955.2
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.98114 -0.54937 -0.08146  0.65884  2.20024
##
## Random effects:
## Groups Name Variance Std.Dev.
## calf_id (Intercept) 20.4 4.516
## Residual 748.7 27.363
## Number of obs: 105, groups: calf_id, 30
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      41.022      7.235   74.132   5.670 2.59e-07 ***
## treatmentpair       1.357      8.686   73.587   0.156  0.876
## day_split5-35     120.315     11.570   78.172  10.399 < 2e-16 ***
## day_split35-63    163.315     11.570   78.172  14.116 < 2e-16 ***
## treatmentpair:day_split5-35   3.835     14.058   76.435   0.273  0.786
## treatmentpair:day_split35-63  4.407     13.906   75.921   0.317  0.752
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr) trtmnt d_5-35 d_35-6 t:_5-3
## treatmentpr -0.833
## dy_splt5-35 -0.602  0.501
## dy_spl35-63 -0.602  0.501  0.378
## trtmn:_5-35  0.495 -0.593 -0.823 -0.311
## trtm:_35-63  0.501 -0.600 -0.315 -0.832  0.372
```

```
# residual diagnostic plot, check assumptions
anova(model_day_treatment_obfeatures)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##              Sum Sq Mean Sq NumDF DenDF F value Pr(>F)
## treatment           328      328      1 28.459  0.4382 0.5133
## day_split          487560  243780      2 74.108 325.5827 <2e-16 ***
## treatment:day_split      96       48      2 74.108  0.0641 0.9379
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# pairwise comparisons
emmeans(model_day_treatment_obfeatures, pairwise ~ treatment|day_split)
```

```

## $emmeans
## day_split = 1-5:
##   treatment  emmean    SE    df lower.CL upper.CL
## individual   41.0  7.27  76.7     26.5     55.5
## pair         42.4  4.82  75.0     32.8     52.0
##
## day_split = 5-35:
##   treatment  emmean    SE    df lower.CL upper.CL
## individual  161.3  9.26  98.8    143.0    179.7
## pair        166.5  6.55  98.8    153.5    179.5
##
## day_split = 35-63:
##   treatment  emmean    SE    df lower.CL upper.CL
## individual  204.3  9.26  98.8    186.0    222.7
## pair        210.1  6.20  98.7    197.8    222.4
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## $contrasts
## day_split = 1-5:
##   contrast      estimate    SE    df t.ratio p.value
## individual - pair    -1.36  8.73  76.2   -0.156  0.8768
##
## day_split = 5-35:
##   contrast      estimate    SE    df t.ratio p.value
## individual - pair    -5.19 11.30  98.8   -0.458  0.6480
##
## day_split = 35-63:
##   contrast      estimate    SE    df t.ratio p.value
## individual - pair    -5.76 11.10  98.8   -0.517  0.6061
##
## Degrees-of-freedom method: kenward-roger

```

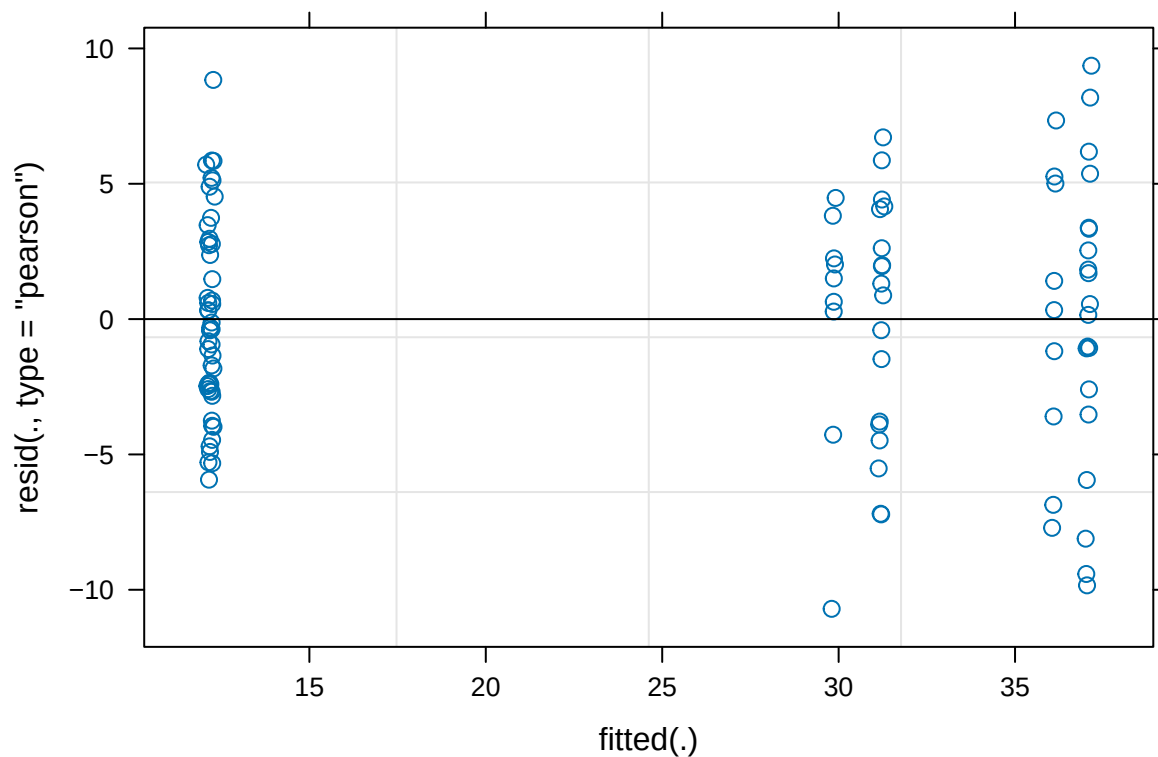
Faith's PD

```

# Build model
model_day_treatment_faithpd <- lmer(faith_pd ~ treatment*day_split + (1|calf_id), data = metadata)

# Residual diagnostic plot, check assumptions
plot(model_day_treatment_faithpd)

```



```
# look at estimated variance components
summary(model_day_treatment_faithpd)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: faith_pd ~ treatment * day_split + (1 | calf_id)
## Data: metadata
##
## REML criterion at convergence: 592
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.42341 -0.60752  0.07115  0.67203  2.11953
##
## Random effects:
## Groups Name Variance Std.Dev.
## calf_id (Intercept) 0.08982 0.2997
## Residual 19.51194 4.4172
## Number of obs: 105, groups: calf_id, 30
##
## Fixed effects:
##              Estimate Std. Error    df t value Pr(>|t|)
## (Intercept) 12.13369 1.14523 75.07412 10.595 < 2e-16 ***
## treatmentpair 0.09934 1.37486 74.71446 0.072 0.943
## day_split5-35 17.72803 1.86339 77.58650 9.514 1.15e-14 ***
## day_split35-63 23.97542 1.86339 77.58650 12.867 < 2e-16 ***
## treatmentpair:day_split5-35 1.25092 2.26513 75.82468 0.552 0.582
## treatmentpair:day_split35-63 0.87113 2.24099 75.27189 0.389 0.699
## ---
```

```

## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##          (Intr) trtmnt d_5-35 d_35-6 t:_5-3
## treatmentpr -0.833
## dy_splt5-35 -0.611  0.509
## dy_spl35-63 -0.611  0.509  0.376
## trtmn:_5-35  0.502 -0.603 -0.823 -0.309
## trtm:_35-63  0.508 -0.609 -0.312 -0.832  0.370

# residual diagnostic plot, check assumptions
anova(model_day_treatment_faithpd)

## Type III Analysis of Variance Table with Satterthwaite's method
##              Sum Sq Mean Sq NumDF  DenDF  F value Pr(>F)
## treatment           13.6    13.6      1 27.690   0.6975 0.4108
## day_split          10760.2  5380.1      2 73.401 275.7333 <2e-16 ***
## treatment:day_split     6.7      3.4      2 73.401   0.1722 0.8422
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

# pairwise comparisons
emmeans(model_day_treatment_faithpd, pairwise ~ treatment|day_split)

## $emmeans
## day_split = 1-5:
##   treatment emmean    SE   df lower.CL upper.CL
## individual  12.1 1.150 78.3     9.84    14.4
## pair       12.2 0.763 77.3    10.71    13.8
##
## day_split = 5-35:
##   treatment emmean    SE   df lower.CL upper.CL
## individual  29.9 1.480 99.0    26.93    32.8
## pair       31.2 1.050 99.0    29.14    33.3
##
## day_split = 35-63:
##   treatment emmean    SE   df lower.CL upper.CL
## individual  36.1 1.480 99.0    33.18    39.0
## pair       37.1 0.990 99.0    35.12    39.0
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## $contrasts
## day_split = 1-5:
##   contrast      estimate    SE df t.ratio p.value
## individual - pair -0.0993 1.38 78  -0.072  0.9429
##
## day_split = 5-35:
##   contrast      estimate    SE df t.ratio p.value
## individual - pair -1.3503 1.81 99  -0.746  0.4575
##
## day_split = 35-63:

```

```
## contrast      estimate    SE df t.ratio p.value
## individual - pair -0.9705 1.78 99  -0.546  0.5866
##
## Degrees-of-freedom method: kenward-roger
```

treatment figure adjusted for time (box plot) (3 timepoint box plot)

```
model_day_treatment_faithpd <- lmer(faith_pd ~ treatment*day_split + (1|calf_id), data = metadata)
pvals <- pairs(
  emmeans(model_day_treatment_faithpd, ~ treatment | day_split)
) |> as.data.frame()
pvals_table <- pvals %>%
  separate(contrast, into = c("group1","group2"), sep = " - ") %>%
  dplyr::mutate(
    p = round(p.value, 4),
    y.position = max(metadata$faith_pd) + 0.1
  )
pvals_table
```

```
##      group1 group2 day_split  estimate      SE      df    t.ratio
## 1 individual  pair      1-5 -0.09934469 1.381399 78.02914 -0.07191598
## 2 individual  pair      5-35 -1.35026840 1.810465 98.97325 -0.74581293
## 3 individual  pair     35-63 -0.97047201 1.778940 98.97176 -0.54553377
##      p.value      p y.position
## 1 0.9428528 0.9429  46.62512
## 2 0.4575481 0.4575  46.62512
## 3 0.5866139 0.5866  46.62512
```

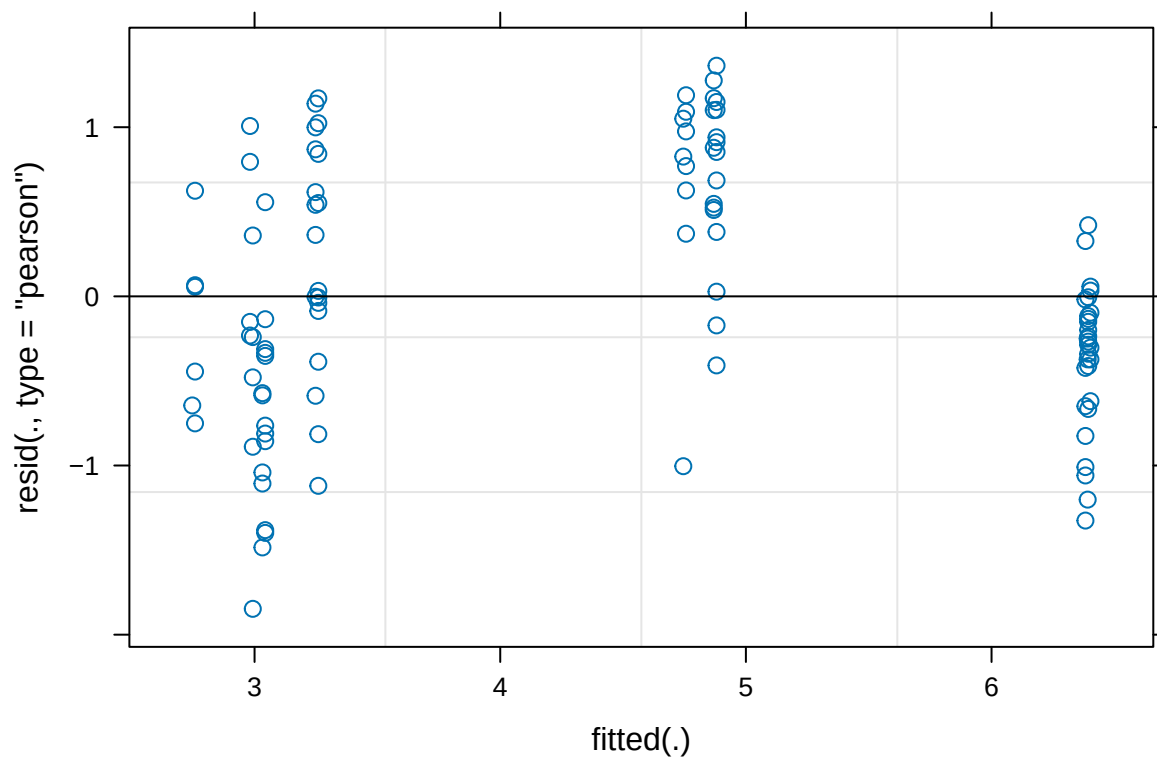
day + facility + sex (question 3)

Shannon

```
# Build model
model_treatment_day_shannon <- lmer(shannon_entropy ~ treatment*day + sex + (1|calf_id), data = metadata)

## boundary (singular) fit: see help('isSingular')

# Residual diagnostic plot, check assumptions
plot(model_treatment_day_shannon)
```



```
# look at estimated variance components
summary(model_treatment_day_shannon)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: shannon_entropy ~ treatment * day + sex + (1 | calf_id)
## Data: metadata
##
## REML criterion at convergence: 262.7
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.4045 -0.6235 -0.1257  0.8126  1.7749
##
## Random effects:
## Groups Name Variance Std.Dev.
## calf_id (Intercept) 0.0000 0.0000
## Residual 0.5902 0.7683
## Number of obs: 105, groups: calf_id, 30
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  2.698879   0.208290 100.000000  12.957 <2e-16 ***
## treatmentpair  0.290432   0.242636 100.000000   1.197  0.234
## day           0.058788   0.005353 100.000000  10.982 <2e-16 ***
## sexmale      -0.011269   0.155407 100.000000  -0.073  0.942
## treatmentpair:day -0.004754  0.006433 100.000000  -0.739  0.462
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Correlation of Fixed Effects:
##          (Intr) trtmnt day    sexmal
## treatmentpr -0.799
## day          -0.725  0.624
## sexmale      -0.245 -0.033 -0.006
## trtmntpr:dy  0.606 -0.745 -0.832 -0.004
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```
# residual diagnostic plot, check assumptions
anova(model_treatment_day_shannon)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##          Sum Sq Mean Sq NumDF DenDF  F value Pr(>F)
## treatment      0.846   0.846     1    100   1.4328 0.2341
## day          181.511 181.511     1    100 307.5375 <2e-16 ***
## sex              0.003   0.003     1    100   0.0053 0.9423
## treatment:day    0.322   0.322     1    100   0.5461 0.4617
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# pairwise comparisons
emmeans(model_treatment_day_shannon, pairwise ~ treatment|day)
```

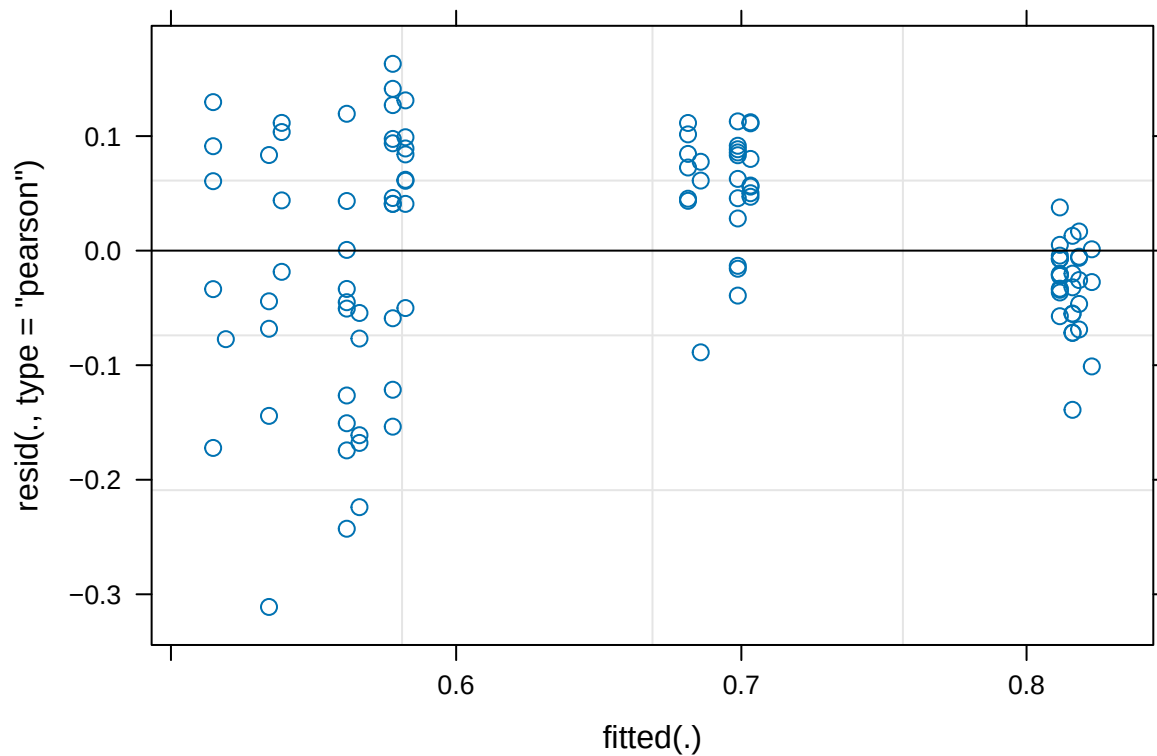
```
## $emmeans
## day = 27.9:
## treatment emmean      SE    df lower.CL upper.CL
## individual  4.34 0.1370 29.5      4.06      4.61
## pair        4.49 0.0924 25.1      4.30      4.68
##
## Results are averaged over the levels of: sex
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## $contrasts
## day = 27.9:
## contrast      estimate      SE    df t.ratio p.value
## individual - pair -0.158 0.162 27.2  -0.972  0.3398
##
## Results are averaged over the levels of: sex
## Degrees-of-freedom method: kenward-roger
```

Evenness

```
# Build model
model_treatment_day_evenness <- lmer(pielou_evenness ~ treatment*day + sex + (1|calf_id), data = metadata)
```

```
## boundary (singular) fit: see help('isSingular')
```

```
# Residual diagnostic plot, check assumptions
plot(model_treatment_day_evenness)
```



```
# look at estimated variance components
summary(model_treatment_day_evenness)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: pielou_evenness ~ treatment * day + sex + (1 | calf_id)
## Data: metadata
##
## REML criterion at convergence: -157.4
##
## Scaled residuals:
##   Min      1Q  Median      3Q      Max
## -3.3079 -0.5380  0.0068  0.8250  1.7344
##
## Random effects:
##   Groups   Name                Variance Std.Dev.
## calf_id   (Intercept)  0.000000  0.00000
## Residual                    0.008842  0.09403
## Number of obs: 105, groups:  calf_id, 30
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  5.099e-01  2.549e-02 1.000e+02  20.000  <2e-16 ***
## treatmentpair 4.772e-02  2.970e-02 1.000e+02   1.607    0.111
## day           4.897e-03  6.552e-04 1.000e+02   7.475   3e-11 ***
```

```
## sexmale          4.449e-03  1.902e-02  1.000e+02  0.234  0.816
## treatmentpair:day -8.648e-04  7.874e-04  1.000e+02  -1.098  0.275
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##          (Intr) trtmnt day    sexmal
## treatmentpr -0.799
## day          -0.725  0.624
## sexmale      -0.245 -0.033 -0.006
## trtmntpr:dy  0.606 -0.745 -0.832 -0.004
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```
# residual diagnostic plot, check assumptions
anova(model_treatment_day_evenness)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##          Sum Sq Mean Sq NumDF DenDF  F value Pr(>F)
## treatment    0.02283  0.02283     1    100   2.5824 0.1112
## day          1.13719  1.13719     1    100 128.6119 <2e-16 ***
## sex           0.00048  0.00048     1    100   0.0547 0.8156
## treatment:day 0.01067  0.01067     1    100   1.2064 0.2747
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

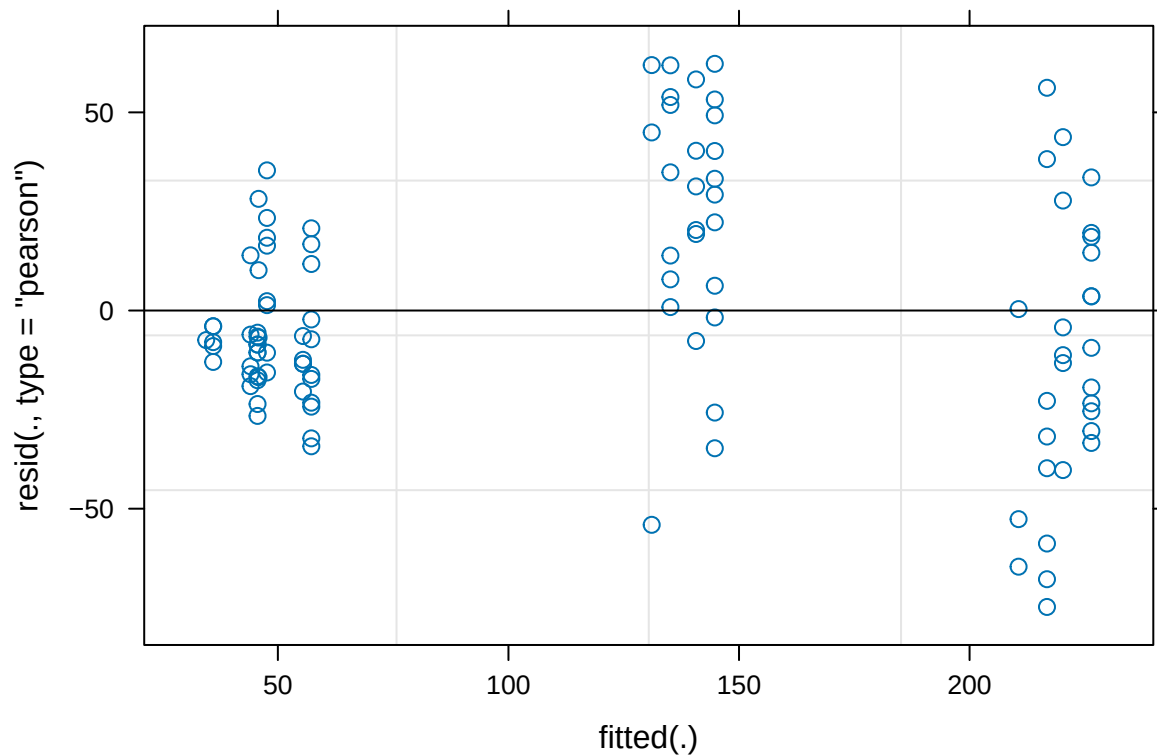
```
# pairwise comparisons
emmeans(model_treatment_day_evenness, pairwise ~ treatment|day)
```

```
## $emmeans
## day = 27.9:
##   treatment emmean      SE    df lower.CL upper.CL
## individual  0.649 0.0167 29.5    0.615    0.683
## pair        0.672 0.0113 25.1    0.649    0.696
##
## Results are averaged over the levels of: sex
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## $contrasts
## day = 27.9:
##   contrast      estimate      SE    df t.ratio p.value
## individual - pair -0.0236 0.0199 27.2   -1.187  0.2456
##
## Results are averaged over the levels of: sex
## Degrees-of-freedom method: kenward-roger
```

Observed features

```
# Build model
model_treatment_day_obfeatures <- lmer(observed_features ~ treatment*day + sex + (1|calf_id), data = me
## boundary (singular) fit: see help('isSingular')
```

```
# Residual diagnostic plot, check assumptions
plot(model_treatment_day_obfeatures)
```



```
# look at estimated variance components
summary(model_treatment_day_obfeatures)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: observed_features ~ treatment * day + sex + (1 | calf_id)
## Data: metadata
##
## REML criterion at convergence: 999.9
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.4405 -0.5630 -0.2101  0.6392  2.0313
##
## Random effects:
## Groups Name Variance Std.Dev.
## calf_id (Intercept) 0.0 0.00
## Residual 939.2 30.65
## Number of obs: 105, groups: calf_id, 30
##
## Fixed effects:
## Estimate Std. Error df t value Pr(>|t|)
## (Intercept) 41.22956 8.30879 100.00000 4.962 2.87e-06 ***
## treatmentpair 1.44204 9.67887 100.00000 0.149 0.882
## day 2.84162 0.21353 100.00000 13.308 < 2e-16 ***
```

```
## sexmale          -9.61866    6.19927 100.00000 -1.552    0.124
## treatmentpair:day 0.07487    0.25662 100.00000  0.292    0.771
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##          (Intr) trtmnt day    sexmal
## treatmentpr -0.799
## day          -0.725  0.624
## sexmale      -0.245 -0.033 -0.006
## trtmntpr:dy  0.606 -0.745 -0.832 -0.004
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```
# residual diagnostic plot, check assumptions
anova(model_treatment_day_obfeatures)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##          Sum Sq Mean Sq NumDF DenDF  F value Pr(>F)
## treatment      21        21      1    100   0.0222 0.8819
## day           472792  472792      1    100 503.4131 <2e-16 ***
## sex            2261      2261      1    100   2.4074 0.1239
## treatment:day    80        80      1    100   0.0851 0.7711
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# pairwise comparisons
emmeans(model_treatment_day_obfeatures, pairwise ~ treatment|day)
```

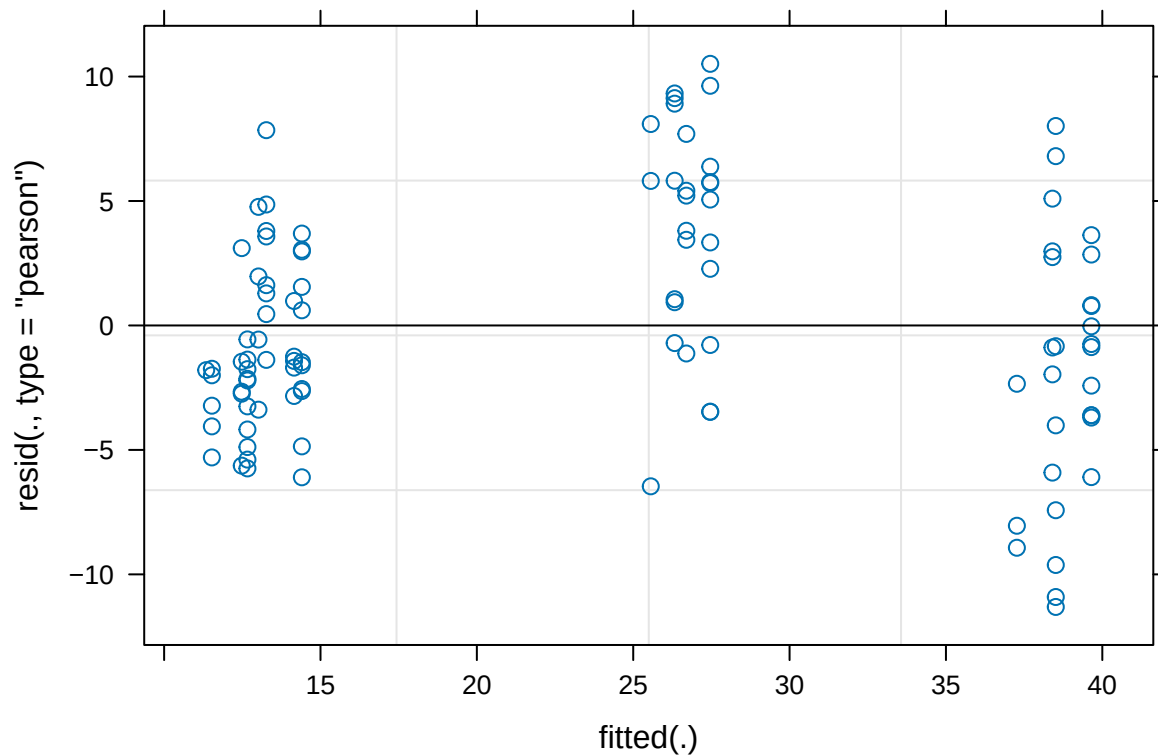
```
## $emmeans
## day = 27.9:
##   treatment emmean   SE    df lower.CL upper.CL
## individual   116 5.45 29.5    105      127
## pair         119 3.69 25.1    112      127
##
## Results are averaged over the levels of: sex
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## $contrasts
## day = 27.9:
##   contrast      estimate    SE    df t.ratio p.value
## individual - pair   -3.53 6.47 27.2  -0.546  0.5895
##
## Results are averaged over the levels of: sex
## Degrees-of-freedom method: kenward-roger
```

Faith's PD

```
# Build model
model_treatment_day_faithpd <- lmer(faith_pd ~ treatment*day + sex + (1|calf_id), data = metadata)

## boundary (singular) fit: see help('isSingular')
```

```
# Residual diagnostic plot, check assumptions
plot(model_treatment_day_faithpd)
```



```
# look at estimated variance components
summary(model_treatment_day_faithpd)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: faith_pd ~ treatment * day + sex + (1 | calf_id)
## Data: metadata
##
## REML criterion at convergence: 630.9
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.3353 -0.6656 -0.1722  0.7107  2.1706
##
## Random effects:
## Groups Name Variance Std.Dev.
## calf_id (Intercept) 0.00 0.000
## Residual 23.45 4.843
## Number of obs: 105, groups: calf_id, 30
##
## Fixed effects:
## Estimate Std. Error df t value Pr(>|t|)
## (Intercept) 12.06397 1.31293 100.00000 9.189 5.94e-15 ***
## treatmentpair 0.16444 1.52942 100.00000 0.108 0.915
## day 0.41813 0.03374 100.00000 12.392 < 2e-16 ***
```

```
## sexmale          -1.13743    0.97959 100.00000 -1.161    0.248
## treatmentpair:day 0.01713    0.04055 100.00000  0.422    0.674
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) trtmnt day    sexmal
## treatmentpr -0.799
## day          -0.725  0.624
## sexmale      -0.245 -0.033 -0.006
## trtmntpr:dy  0.606 -0.745 -0.832 -0.004
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```
# residual diagnostic plot, check assumptions
anova(model_treatment_day_faithpd)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##              Sum Sq Mean Sq NumDF DenDF  F value Pr(>F)
## treatment          0.3      0.3      1    100   0.0116 0.9146
## day             10384.9 10384.9      1    100 442.8429 <2e-16 ***
## sex                31.6      31.6      1    100   1.3482 0.2484
## treatment:day       4.2       4.2      1    100   0.1784 0.6736
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# pairwise comparisons
emmeans(model_treatment_day_faithpd, pairwise ~ treatment|day)
```

```
## $emmeans
## day = 27.9:
##   treatment emmean    SE    df lower.CL upper.CL
## individual  23.2 0.862 29.5     21.4     24.9
## pair        23.8 0.582 25.1     22.6     25.0
##
## Results are averaged over the levels of: sex
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## $contrasts
## day = 27.9:
##   contrast      estimate    SE    df t.ratio p.value
## individual - pair  -0.643 1.02 27.2  -0.629  0.5348
##
## Results are averaged over the levels of: sex
## Degrees-of-freedom method: kenward-roger
```

Sample type figure

```
model_treatment_day_obfeatures <- lmer(observed_features ~ treatment*day + sex + (1|calf_id), data = me
```

```
## boundary (singular) fit: see help('isSingular')
```

```
pvals <- pairs(  
  emmeans(model_treatment_day_obfeatures, ~ treatment | day)  
) |> as.data.frame()  
  
pvals_table <- pvals %>%  
  separate(contrast, into = c("group1","group2"), sep = " - ") %>%  
  dplyr::mutate(  
    p = round(p.value, 4),  
    y.position = max(metadata$observed_features) + 0.1  
  )  
pvals_table
```

```
##      group1 group2      day estimate      SE      df      t.ratio  p.value  
## 1 individual  pair 27.93333 -3.533281 6.471739 27.21044 -0.5459555 0.5895451  
##      p y.position  
## 1 0.5895      273.1
```